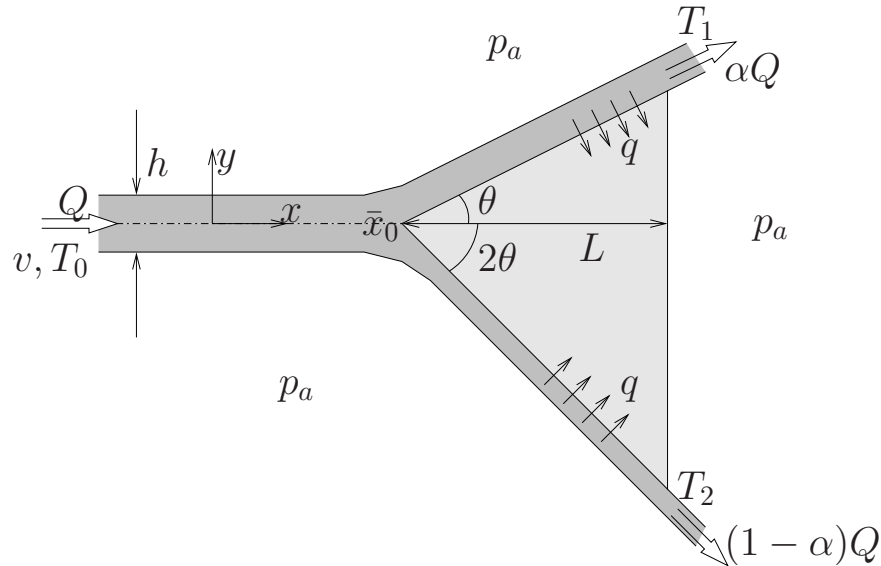


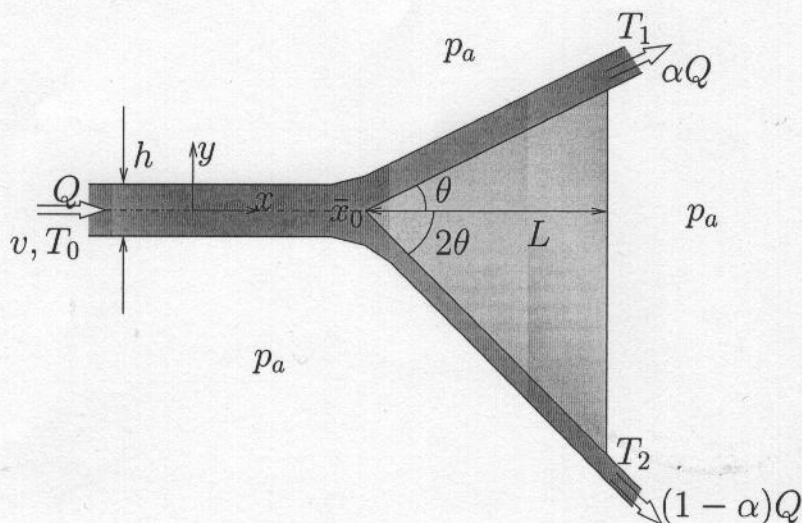
PROBLEM 1 (1 hour)

A plane liquid jet of constant density ρ , thickness h and uniform velocity v , impinges on a non-symmetric wedge of geometry and dimensions shown in the figure. The wedge splits the jet into two layers of unequal flux, such that a portion α of the volume flux is deflected an angle θ , and the rest of the fluid is deflected an angle 2θ .



- The flow is steady.
 - Viscous and gravitational forces are negligible.
 - The known data are $(\rho, c, v, h, \alpha, \theta, L, p_a, T_0, T_1, T_2)$.
1. (1 point). Noting that the conditions for the use of Bernoulli's theorem are met, determine the (uniform) velocity of the fluid on each deflected layer.
 2. (2 points). Determine the thickness h_1 and h_2 of each layer.
 3. (2.5 points). Determine the force $\bar{F}$ (per unit length in the z -direction) that the liquid jet and the ambient air exert on the wedge.
 4. (2.5 points). Determine the moment $\bar{M}_0$ due to the force $\bar{F}$ about the corner $\bar{x}_0$ of the wedge. Find the value of the fraction α that would produce a null moment, $\bar{M}_0 = 0$.
 5. (2 points). Knowing the uniform temperature T_0 of the incoming jet, the uniform temperatures T_1 and T_2 of the two liquid layers when they lose contact with the wedge, and the constant specific heat c of the liquid, determine the heat q transferred per unit surface from the liquid to the wedge. Assume that the heat flux is uniform on the wedge surface and that there is no heat transfer by conduction between the liquid and the ambient air.

A plane liquid jet of thickness h and velocity v impinges on a non-symmetric wedge of geometry and dimensions shown in the figure. The wedge splits the jet into two layers of unequal flux, as shown in the figure. A portion α of the volume flux is deflected an angle θ and the rest of the fluid is deflected an angle 2θ . The flow is steady and viscous forces and gravity effects are negligible. The density ρ of the liquid is known and constant.



- Noting that the conditions for the use of Bernoulli's theorem are met, determine the (uniform) velocity of the fluid on each deflected layer. $v_1 = v_2 = v$ (1)
- Determine the thickness h_1 and h_2 of each layer. $h_1 = \alpha h$
 $h_2 = (1-\alpha)h$ (2)
- Determine the force (per unit length in the z -direction) $\bar{F}$ that the liquid jet and the ambient air exert on the wedge.
- Determine the moment $\bar{M}_0$ due to the force $\bar{F}$ about the corner $\bar{x}_0$ of the wedge. Find the value of the fraction α that would produce a null moment, $\bar{M}_0 = 0$.
- Knowing the temperature T_0 of the incoming jet and the temperatures T_1 and T_2 of the two liquid layers when they lose contact with the wedge, determine the heat q transferred per unit surface from the liquid to the wedge. Assume that the heat flux is uniform on the wedge surface and that there is no heat transfer by conduction between the liquid and the ambient air. The specific heat c of the liquid is known and constant.

$$\bar{F} = \rho v^2 h \left\{ (1-\alpha) \sin \theta - (1-\alpha) \sin 2\theta \right\} \bar{e}_x - \left\{ \alpha \sin \theta - (1-\alpha) \sin 2\theta \right\} \bar{e}_y \quad (2.5)$$

$$\bar{M}_0 = \rho v^2 \frac{h^2}{2} (2\alpha - 1) \bar{e}_z \quad (2) \quad \alpha = \frac{1}{2} \text{ for } \bar{M}_0 = 0 \quad (0.5)$$

$$q = \frac{\rho c v h}{\frac{L}{\cos \theta} + \frac{L}{\cos 2\theta}} (T_0 - \alpha T_1 - (1-\alpha) T_2) \quad (2)$$